

Ex.No: 10

Installation and Configuration of Firewall Using iptables

AIM:

ALGORITHM:

CODE:

```
sudo iptables -L -v -n
```

```
sudo iptables -F
```

```
sudo iptables -X
```

```
sudo iptables -A INPUT -i lo -j ACCEPT
```

```
sudo iptables -A INPUT -m conntrack --ctstate RELATED,ESTABLISHED -j ACCEPT
```

```
sudo iptables -A INPUT -p tcp --dport 22 -j ACCEPT
```

```
sudo iptables -A INPUT -p tcp --dport 80 -j ACCEPT
```

```
sudo iptables -A INPUT -p tcp --dport 8080 -j ACCEPT
```

```
sudo iptables -A INPUT -p tcp --dport 9090 -j DROP
```

```
sudo iptables -P INPUT DROP
```

```
sudo iptables -P FORWARD DROP
```

```
sudo iptables -P OUTPUT ACCEPT
```

```
sudo iptables -L -v -n
```

OUTPUT:

```
acerthaniya@DESKTOP-QMTI2UM:~$ sudo iptables -L -v -n
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target    prot opt in     out     source                   destination

Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target    prot opt in     out     source                   destination

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target    prot opt in     out     source                   destination
acerthaniya@DESKTOP-QMTI2UM:~$ sudo iptables -F
acerthaniya@DESKTOP-QMTI2UM:~$ sudo iptables -X
acerthaniya@DESKTOP-QMTI2UM:~$ sudo iptables -A INPUT -i lo -j ACCEPT
acerthaniya@DESKTOP-QMTI2UM:~$ sudo iptables -A INPUT -m conntrack --ctstate RELATED,ESTABLISHED -j ACCEPT
acerthaniya@DESKTOP-QMTI2UM:~$ sudo iptables -A INPUT -p tcp --dport 22 -j ACCEPT
acerthaniya@DESKTOP-QMTI2UM:~$ sudo iptables -A INPUT -p tcp --dport 80 -j ACCEPT
acerthaniya@DESKTOP-QMTI2UM:~$ sudo iptables -A INPUT -p tcp --dport 8080 -j ACCEPT
acerthaniya@DESKTOP-QMTI2UM:~$ sudo iptables -A INPUT -p tcp --dport 9090 -j DROP
acerthaniya@DESKTOP-QMTI2UM:~$ sudo iptables -P INPUT DROP
acerthaniya@DESKTOP-QMTI2UM:~$ sudo iptables -P FORWARD DROP
acerthaniya@DESKTOP-QMTI2UM:~$ sudo iptables -P OUTPUT ACCEPT
acerthaniya@DESKTOP-QMTI2UM:~$ sudo iptables -L -v -n
Chain INPUT (policy DROP 0 packets, 0 bytes)
 pkts bytes target    prot opt in     out     source                   destination
 0      0 ACCEPT    all  --  lo     *       0.0.0.0/0               0.0.0.0/0
10    2160 ACCEPT    all  --  *      *       0.0.0.0/0               0.0.0.0/0               ctstate RELATED,ESTABLISHED
 0      0 ACCEPT    tcp  --  *      *       0.0.0.0/0               0.0.0.0/0               tcp dpt:22
 0      0 ACCEPT    tcp  --  *      *       0.0.0.0/0               0.0.0.0/0               tcp dpt:80
 0      0 ACCEPT    tcp  --  *      *       0.0.0.0/0               0.0.0.0/0               tcp dpt:8080
 0      0 DROP      tcp  --  *      *       0.0.0.0/0               0.0.0.0/0               tcp dpt:9090

Chain FORWARD (policy DROP 0 packets, 0 bytes)
 pkts bytes target    prot opt in     out     source                   destination

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target    prot opt in     out     source                   destination
acerthaniya@DESKTOP-QMTI2UM:~$ python3 -m http.server 8080
Serving HTTP on 0.0.0.0 port 8080 (http://0.0.0.0:8080/) ...
127.0.0.1 - - [09/Sep/2026 02:35:10] "GET / HTTP/1.1" 200 -
```

```
acerthaniya@DESKTOP-QMTI2UM:/mnt/c/WINDOWS/system32$ sudo iptables -L INPUT -v -n
[sudo: authenticate] Password:
Chain INPUT (policy DROP 0 packets, 0 bytes)
 pkts bytes target    prot opt in     out     source                   destination
40    272K ACCEPT    all  --  lo     *       0.0.0.0/0               0.0.0.0/0
68    14688 ACCEPT    all  --  *      *       0.0.0.0/0               0.0.0.0/0               ctstate RELATED,ESTABLISHED
 0      0 ACCEPT    tcp  --  *      *       0.0.0.0/0               0.0.0.0/0               tcp dpt:22
 0      0 ACCEPT    tcp  --  *      *       0.0.0.0/0               0.0.0.0/0               tcp dpt:80
 0      0 ACCEPT    tcp  --  *      *       0.0.0.0/0               0.0.0.0/0               tcp dpt:8080
 0      0 DROP      tcp  --  *      *       0.0.0.0/0               0.0.0.0/0               tcp dpt:9090
```

RESULT: